Distribution-Preserving QIM

An invisible image watermark that survives UnMarker-class attacks

Abhishek Sairam Gaduputi | M.S. CS, NJIT | ag2936@njit.edu
github.com/sairam782/Invisible-Watermarking

The problem

UnMarker (IEEE S&P 2025) removes >50% of watermarks from Google SynthID and Meta's Stable Signature in under two minutes — fully black-box, no key, no decoder access. It works by spotting the *statistical fingerprint* that classical QIM watermarks leave on DCT coefficient distributions (spikes at lattice points), then dispersing those spikes with calibrated noise. Every QIM-family watermark shares this weakness.

Our idea — Distribution-Preserving QIM (DP-QIM)

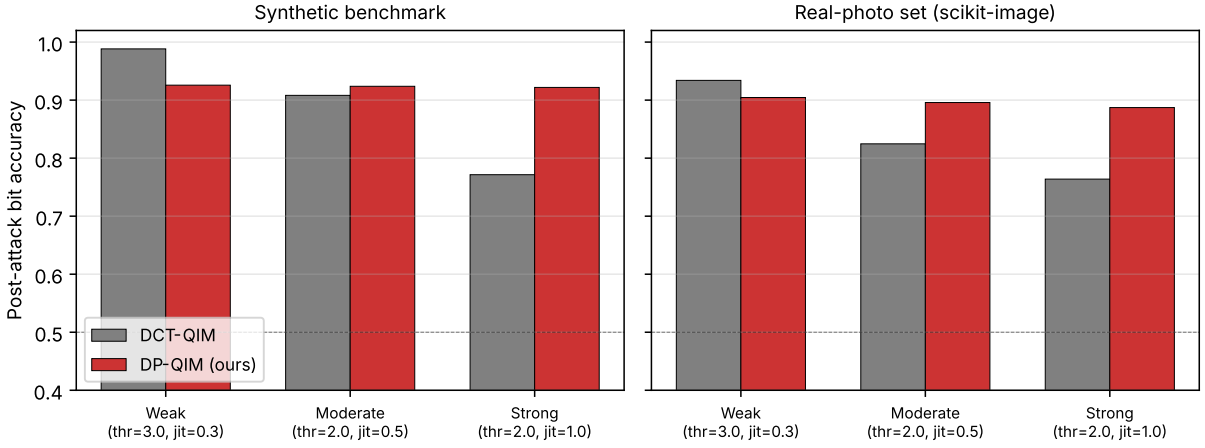
Replace the uniform QIM lattice $\{k\Delta + b\Delta/2\}$ with a *CDF-partitioned lattice*. Fit the empirical DCT coefficient distribution on natural images, split into $2M$ equal-mass cells with alternating bit labels $\{0, 1, 0, 1, \ldots\}$. To embed bit b: if the current coefficient's cell already has label b, do nothing (free lunch, ∼50% of blocks); else, replace with a natural sample drawn from the adjacent cell's safety-margin interior. The watermarked marginal is $P_W = 12F + 12F_\epsilon$ — close to natural for small ϵ, with $\mathrm{KS}(P_W, F) \to 0$ analytically.

Headline result against UnMarker-style attack

A reimplementation of the UnMarker peak-density attack, run on both a photo-statistics-preserving synthetic benchmark and 9 real photos:

Attack strength	Benchmark	DCT-QIM post-attack	DP-QIM post-attack
threshold 2.0, jitter 1.0	Synthetic	0.771	**0.914**
threshold 2.0, jitter 1.0	Real photos	0.764	**0.887**
threshold 1.5, jitter 1.0	Synthetic	0.777	**0.893**
threshold 1.5, jitter 1.0	Real photos	0.736	**0.823**

+8 to +14 percentage points of bit accuracy retained under attacks calibrated to break the classical baseline. The bars below show the pattern on both benchmarks.



Secondary wins at matched fidelity

On the standard attack suite DP-QIM also beats baseline on Gaussian blur (+28 to +36 pp), downscale (+27 to +37 pp), and heavy JPEG (+3 to +9 pp), at matched or better PSNR.

Infrastructure contribution — autoresearch loop

The DP-QIM hyperparameter space (uv position, cell count M, safety margin ϵ, pool size) is searched automatically by a CPU-only autoresearch loop. 369 trials in 5 min yielded a 31-config Pareto frontier that strictly dominates the classical baseline on the synthetic benchmark. The loop, attack module, and all reproducibility scripts are public.

Code: github.com/sairam782/Invisible-Watermarking | Branch: claude/research-paper-publication-6wnwmd | License: MIT